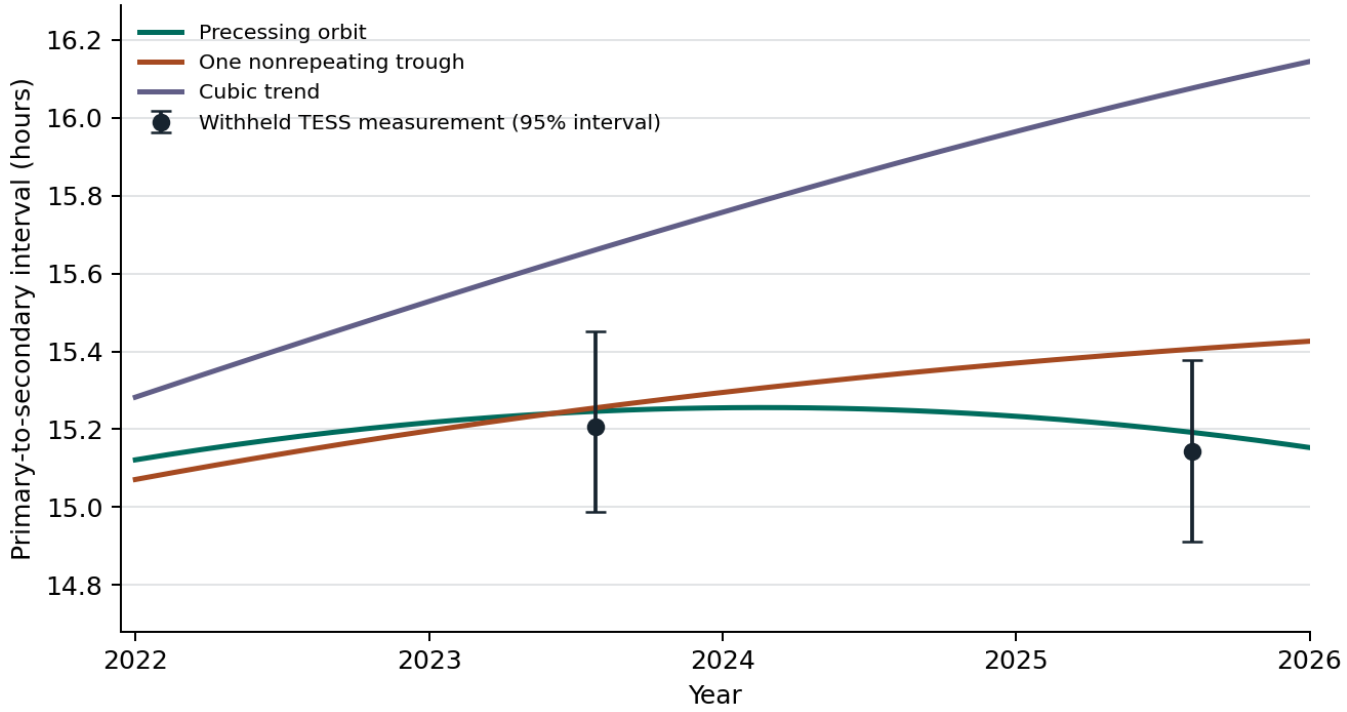
Testing a 28-year orbit against later eclipses

The two stars orbit every 28.07 hours. From 2010–2014 to 2022–2024, the interval from the primary to the following secondary eclipse grew by 2.21 hours. An elliptical orbit that slowly turns would produce a change like this. We tested it against observations taken after a fixed time cutoff.



Absolute errors on withheld TESS measurements

Model fitted before 2022	2023 error	2025 error
Precessing orbit	2.4 min	3.0 min
One-time dip followed by a plateau	2.9 min	15.8 min
Cubic timing trend	27.3 min	56.0 min

This retrospective test withheld the last 96 OGLE measurements and both of the later TESS groups. The earlier OGLE, Gaia and TESS data determined all fitted parameters. TESS measured 15.21 hours in 2023 and 15.14 in 2025. The timing precision at either epoch is about seven minutes (one bootstrap standard deviation).

One-time changes remain plausible. In 102 of 300 simulations generated by the plateau model, the precessing model gained at least as much predictive advantage as in the real record. A second reversal of the interval would provide a more decisive test.

Ultraviolet eclipses and comparable binaries

Swift saw both eclipses in the ultraviolet

Swift's ultraviolet telescope recorded both eclipses. At times predicted from OGLE, the target's brightness relative to a nearby reference star fell by 41% ± 3% at primary eclipse and 32% ± 5% at secondary eclipse. The analysis used 275 accepted image measurements over 267 nights. Errors are one standard error grouped by night.

Swift independently measures the interval change

Fitting both ultraviolet eclipse centers gives 14.66 hours in 2016–2020 and 15.49 hours in 2021–2026. The increase is 50 minutes (conditional 95% night-bootstrap interval 2–116 minutes). Changing the aperture gives increases of 41–50 minutes. All three accepted 2026 observations fell outside both eclipse cores, so the late value describes several years.

XMM-Newton caught the primary eclipse in 2015

On March 24, an ultraviolet exposure covering the predicted primary minimum measured 0.428 ± 0.056 magnitudes of dimming relative to the surrounding two exposures. Of 75 comparison sources of similar brightness, none had a dip this large. The exposure lasted 67 minutes and cannot establish the exact minimum time.

How common was a comparable reversal?

A catalog-based search selected 676 non-contact binaries of similar period, brightness and catalog eclipse depth. Of the 442 with adequate timing in four independent date ranges, four other binaries matched or exceeded this target's two changes in opposite directions. The comparison recovered OGLE-SMC-ECL-03825, which OGLE had already flagged for visible apsidal motion.

Why it merits an individual study

Hong and colleagues measured apsidal periods in 90 SMC systems drawn from a sample brighter than I = 18.0 mag. Our binary was fainter, at I = 18.232 mag. Only one of their 90 had a shorter orbital period. This comparison describes a selected catalog and does not establish a record. Searches under the catalog identifiers found no prior target-specific report of this reversal.

The archive contains the comparison-star measurements, catalog crossmatches, ultraviolet exposure dates, fluxes, and code. The target was chosen after a larger exploratory search, so 4 of 441 comparison stars cannot be interpreted as a false-alarm probability.

Swift archive XMM-Newton archive OGLE-III catalog Hong et al. (2016)

A test that can distinguish the explanations

The physical fit

The joint OGLE, Gaia and TESS timing fit gives a precession period of 28.25 years. The conditional 95% interval is 27.59–28.87 years. A separate fit that projects two stellar disks along a Kepler orbit and calculates their changing overlap gives 28.45 years and eccentricity about 0.067. Both calculations reuse OGLE observations. The data span less than one modeled cycle.

Where the geometry is uncertain

The inclination depends on how much light comes from unresolved sources. TESS suggests the secondary eclipse lasted longer than the primary in 2025, but its fitted width difference varies from about 8 to 22 minutes with the aperture or sky reduction. Resampling single orbits also allows the opposite ordering. The weak brightness modulation outside eclipse puts pressure on a simple equilibrium-tide estimate.

Full-data forecasts for the primary-to-secondary gap

Date	Precessing orbit	One-time dip	Difference
July 1, 2027	14.92 h	15.32 h	24 min
July 1, 2028	14.72 h	15.34 h	37 min
July 1, 2030	14.22 h	15.35 h	68 min

Two observing sites can time the same orbit

An example is October 17–18, 2027. Siding Spring, Australia, can cover the primary on October 17 from 11:06–15:25 UTC. Cerro Tololo, Chile, can cover the following secondary on October 18 from 02:07–06:26 UTC. The star stays above 43° on the sampled visibility grid. The two models differ by 28 minutes for this pair.

Timing several pairs and measuring the full duration of both eclipses would test repeatability as well as the orientation of the orbit. The accompanying file provides 17 geometrically feasible pairs and the assumptions used to calculate them.

OGLE-IV photometry MAST / TESS archive Siding Spring